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Lab 05: Rollback

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CI/CD ENGINEERING · KNOWLEDGE BASE

Lab 04: Deployment

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Objective

Promote Lab 03's digest through staging to production with an explicit approval gate — same digest, two environments.

Prerequisites

- Lab 03 digest in GHCR; concepts: Continuous Delivery, Environments & Release.

Architecture

main → deploy digest to staging (auto) → smoke check → GitHub Environment production (required reviewer) → deploy same digest → verify.

Setup

Repo Settings → Environments → create staging and production; set a required reviewer on production.

Step 1 — Staging auto-deploys

Execute

```
deploy-staging: needs: image environment: staging runs-on: ubuntu-latest steps: - run: echo "deploy ${{ needs.image.outputs.digest }} to staging"
```

Expected Result

Every main merge deploys staging without human action.

Why

Staging proves the artifact continuously; automation here is safe because users never see it.

Step 2 — Production needs a human

Execute

Add deploy-prod with environment: production and needs: [deploy-staging]; merge, then approve the pending deployment.

Expected Result

Production waits for approval; after approval it runs the same digest.

Why

This is Continuous Delivery's one gate: releasable always, deployed by decision (see delivery vs deployment).

Verification

- [] Staging deploys on every merge, no clicks.
- [] Production blocks until approved.
- [] Both environments ran the identical digest.

Failure Scenarios

- Approval never arrives: check environment reviewers and notifications.
- Digests differ between envs: promotion passed a tag, not a digest — pass `outputs.digest` explicitly.

Cleanup

Keep environments; Lab 05 breaks production on purpose.

What This Proved

Promotion paths, environment gates, digest discipline. Continue with Lab 05.

Interview Notes

- What the production approval gate actually blocks — and where reviewers are configured.
- Tag vs digest promotion: how you prove both environments ran identical bytes.

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